# SHUBHAJIT NANDI

## Senior Software Engineer | Backend Systems & Cloud Delivery

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### SUMMARY

Senior software engineer with 11+ years building production systems across telecom, cybersecurity, and voice AI. Lead a five-engineer SaaS team with ownership of backend architecture and cloud delivery. Built a messaging platform handling 3,000 TPS with 90% lower per-message routing costs. Combine Java/Spring Boot development with frontend mentoring, AI integration, and production operations.

### TECHNICAL SKILLS

**Languages & Backend:** Java, Python, C, JavaScript, Spring Boot, Spring Cloud, Hibernate, FastAPI, REST APIs

**Messaging & Architecture:** Apache Kafka, Kafka Connect, NATS JetStream, AWS SQS, microservices, event-driven systems, CQRS, Axon Framework

**Databases:** PostgreSQL, MySQL, MongoDB replica sets, Redis

**Cloud & DevOps:** AWS, Terraform, Docker, Docker Compose, Kubernetes, GitHub Actions, CI/CD, Amazon S3, CloudFront, Nginx

**Performance & Operations:** Grafana k6, load testing, Grafana, logging, tracing, alerting, incident response

**AI & Cross-Stack Integration:** OpenAI and Gemini APIs, structured outputs, tool calling, micro-frontends, frontend mentoring

### PROFESSIONAL EXPERIENCE

#### Aspire Tech Services and Solutions Corp. (US-based client)

Senior Software Engineer | Jun 2025 - Present | Remote, Dhaka

- Lead five engineers delivering ASAT-V2, a security-awareness SaaS platform, owning backend architecture, AWS infrastructure, and production delivery.
- Reduced JavaScript asset loading time by approximately 50% by moving asset delivery to Amazon S3 and CloudFront.
- Built Spring Boot services for authentication, phishing simulations, billing, notifications, and reporting; integrated Stripe and PayPal with coupon, VAT, and discount calculations.
- Provisioned AWS infrastructure with Terraform and automated container builds, deployment, and versioned rollback through GitHub Actions; prepare servers for new deployment environments.
- Integrated OpenAI and Gemini for customer-specific simulation content, implementing structured outputs, tool calling, timeouts, retries, and rate-limit handling.
- Mentored frontend engineers on micro-frontend design and implementation, aligning backend APIs and UI workflows across phishing, billing, and training modules.
- Configured a three-member MongoDB replica set with automatic failover for the SaaS platform.
- Use Grafana k6 for load testing and performance analysis; maintain Grafana dashboards, logging, alerts, and production incident response.

**Verbex**

Senior Software Engineer | Aug 2023 - Jun 2025 | Dhaka, Bangladesh

- Built a Kafka and Kafka Connect data-warehousing pipeline, reducing processing latency by approximately 50% for downstream analytics.
- Developed a custom Kafka Connect transformation plugin to reshape complex source data for BI dashboards.
- Built seven Spring Boot services for a conversational voice AI platform using Java 21, Axon Framework, and CQRS, covering agent management, call orchestration, and transcription analysis.
- Integrated LiveKit SIP trunking to support inbound and outbound calls for the conversational AI platform.
- Implemented GitHub Actions pipelines with Docker and Kubernetes for containerized service deployments.

**Dotlines Bangladesh**

Senior Software Engineer | Nov 2020 - Aug 2023 | Dhaka, Bangladesh

- Built a C-based SMSC using SMPP, SS7, and SCTP, handling 3,000 TPS and reducing per-message routing costs by 90% compared with third-party providers.
- Developed Java payment-gateway services deployed on Grameenphone and Banglalink platforms, integrating multiple third-party payment providers.
- Developed an asynchronous outbound-call gateway using Python, FastAPI, asyncio, and NATS JetStream to increase parallel call-processing capacity.

**SSD-TECH**

Software Engineer | Feb 2015 - Nov 2020 | Dhaka, Bangladesh

- Developed SMS delivery systems in C and PHP with MySQL and Redis, running on Banglalink's production platform at 2,000 TPS.
- Built reusable SMPP and SIP libraries in C for multiple telecom projects; implemented call-patching and conference-call features using Asterisk.

## EDUCATION

Bachelor's in Computer Science and Engineering | 2010 - 2014

Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh

## CERTIFICATION & LANGUAGES

Splunk Certified Cybersecurity Defense Analyst

Bengali: Native | English: Fluent